

ENSO and the 2026-27 harvests

Observations through September 30, 2026. ENSO scenarios from the September 2026 outlook. This bulletin summarizes the frozen atlas release; it does not incorporate a later forecast issue.

We find different crop responses across countries and substantial uncertainty within each country. The central ENSO scenario implies a yield increase for US and Argentine maize and a decrease for Indian Kharif rice and Brazilian maize in the selected panels. Observed weather can point in another direction: the US weather estimate is -3.1%, while the ENSO scenario is +8.1%. The methods cover different sets of reporting units.

ENSO tracking at the same point in earlier events

The September 2026 monthly Niño 3.4 anomaly is 2.86 °C. Removing the tropical mean anomaly gives a relative Niño 3.4 value of 1.97 °C. We compare September with September in earlier events using the same index definitions. These monthly values do not determine the coming winter’s peak.

September	Niño 3.4	Relative Niño 3.4
2026	+2.86 °C	+1.97 °C
2023	+1.60 °C	+0.98 °C
2015	+2.20 °C	+1.65 °C
1997	+2.12 °C	+1.90 °C
1982	+1.49 °C	+1.59 °C

The stored outlook uses lower, central and higher winter ENSO paths. Their relative-index DJF means are 1.26, 1.93 and 2.60 °C. We apply those paths to each crop’s growing season, so timing matters alongside event strength.

Conditional yield changes relative to trend

Crop and harvest	Lower ENSO	Central	Higher ENSO	Central coefficient bounds	Weather so far
United States maize 2026; in the ground	+7.0%	+8.1%	+9.1%	-7.1% to +26.5%	-3.1% Meets checks
India rice (milled), Kharif 2026; in the ground	-3.3%	-3.9%	-4.5%	-16.7% to +12.6%	-2.6% Meets checks
Brazil maize 2027; not planted	-3.3%	-4.3%	-4.8%	-25.2% to +26.7%	-22.6% Checks not met
Argentina maize 2027; not planted	+14.8%	+23.9%	+34.0%	-6.5% to +66.2%	Unavailable Not scored

Negative values denote losses relative to the fitted yield trend. Lower and higher refer to ENSO strength, not loss percentiles. Bounds average local 95% coefficient limits. They are not calibrated national 95% intervals; other harvest drivers add uncertainty.

Implications for monitoring

We would follow the US maize weather estimate alongside local crop reports because the weather estimate meets the release's seasonal coverage and historical-skill checks. The positive ENSO scenario is a historical conditional relationship. The negative weather estimate reflects the conditions observed so far. Neither number establishes the final yield outcome.

Indian Kharif rice has a negative central ENSO estimate and a negative weather estimate, but the scenario interval includes gains. The district maps show geographic differences that the national average obscures. Stable merged HarvestStat districts preserve comparability through boundary changes.

The Brazilian maize weather estimate has only 0.14% production coverage and fails the historical-skill check. We would not treat its large negative number as an actionable national outlook. The Argentine maize panel is not yet planted; its conditional ENSO estimate provides a scenario to track as weather observations accumulate.

From harvest risk to food security

Local yield departures can affect farm incomes, procurement and regional trade even when gains elsewhere offset losses in broader totals. Stocks, prices, trade restrictions, incomes and conflict also affect food access. The atlas does not translate a yield estimate into a population needing assistance.

FEWS NET's October 6 report identifies particular concern in Southern Africa and East African areas with one main rainy season. FEWS NET also distinguishes rainfall and harvest impacts from later lean-season food-security effects. We would use that regional assessment to interpret the atlas's crop-specific estimates and calendars.

How to use the release

Choose a crop on the atlas landing page and open a named country and season. Read the uncertainty and reliability checks beside the estimate, then inspect local coverage. The weather view places observed rainfall and temperature beside their historical ENSO expectations. Yield history places the crop's ENSO exposure beside observed yields and trend departures.

Copy a link to preserve the release and selection. Download a table or figure with its provenance. For a country briefing, open the relevant season and use "Print / save country brief." Changes since the preceding build are listed on the landing page; changes in methods and coverage can move an estimate without new weather observations.

Citation: ENSO yield atlas, release 2026-10-08. Sources and downloadable numerical results: atlas methods and data. Prepared for review and discussion using the frozen September observation cutoff.

Appendix: data and methods

We prioritize HarvestStat v0.1 and v0.2, then Hultgren replication data with agency extensions, then FAOSTAT national records. India uses merged HarvestStat districts. Detailed Hultgren crop calendars, FAO/GIEWS and RiceAtlas provide growing windows where available; each unit page identifies the actual source. We preserve source crop types and keep overlapping seasonal and Annual panels separate.

We fit long-term yield trends and estimate yield relationships with standard and relative Niño 3.4. The analysis selects a unit's own fit where its record is sufficiently long, with regional or country fallbacks for short records. Country and regional regressions are separate estimates, so they need not match an average of the local map. We retain those differences and show the estimator level.

ENSO scenarios combine the observed index with lower, central and higher outlook paths. We apply the paths over each crop's growing window and weight the conditional changes using the analysis release's production weights. HarvestStat production shares and Hultgren cropped-area shares allocate FAOSTAT national totals; area-based allocations are proxies for local production. Temperature and rainfall summaries use BEST and CHIRPS. The weather regression uses observations from the elapsed part of each window. Historical tests use rolling origins and exclude the target harvest year and its neighbors from training.

Weather checks require enough of the growing season, sufficient production coverage, bounded estimation error and positive historical skill at the current lead. Missing skill and negative skill have different meanings. The index-outlook check assumes the future index is known; it does not establish operational forecast skill. A matched-footprint comparison uses identical production weights for both sets of unit estimates and appears separately from the fitted country weather estimate. We do not assign a new aggregate confidence interval to that descriptive comparison.

We shade historical yields using ENSO exposure matched to the exact series and harvest year, including crops spanning calendar years. We retain QC flags, duplicate source observations and missing years for inspection. Full source hashes, methods, build revisions and numerical validation are available with the downloadable release.